

## KING HUSSEIN AB, Jordan

WMO: 402650

Lat: 32.356N

Lon: 36.259E

Elev: 2240

StdP: 13.54

Time Zone: 2.00 (JOR)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 32.4       | 34.8 | 11.0                          | 10.5 | 52.5 | 16.6 | 13.8 | 55.7 | 28.0                  | 48.1 | 22.9 | 48.5 | 2.1                   | 280  | 0.421 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 26.5                   | 98.3            | 66.3 | 95.4 | 65.9 | 92.9 | 65.3 | 72.8                | 84.9 | 71.3 | 83.9 | 70.0 | 82.2 | 8.7                  | 300  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 70.0                            | 119.9 | 75.8 | 68.2 | 112.7 | 74.4 | 66.6 | 106.5 | 73.7 | 38.0          | 84.1 | 36.6 | 83.8 | 35.4 | 82.2 | 81.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |       |                    | n-Year Return Period Values of Extreme Temperature |           |       |            |       |            |       |            |       |
|-------------------|------|------|----------------------------|------|-------|--------------------|----------------------------------------------------|-----------|-------|------------|-------|------------|-------|------------|-------|
| 1%                |      | 2.5% | 5%                         | Mean |       | Standard Deviation |                                                    | n=5 years |       | n=10 years |       | n=20 years |       | n=50 years |       |
| Min               | Max  | Min  | Max                        | Min  | Max   | Min                | Max                                                | Min       | Max   | Min        | Max   | Min        | Max   | Min        | Max   |
| (a)               | (b)  | (c)  |                            | (e)  | (f)   | (g)                | (h)                                                | (i)       | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |
| 21.4              | 18.3 | 15.9 | DB                         | 27.5 | 104.4 | 3.4                | 2.7                                                | 25.1      | 106.4 | 23.1       | 108.0 | 21.2       | 109.6 | 18.7       | 111.6 |
|                   |      |      | WB                         | 25.1 | 75.2  | 4.0                | 2.5                                                | 22.2      | 77.0  | 19.8       | 78.4  | 17.5       | 79.8  | 14.6       | 81.6  |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 64.0  | 46.3  | 49.5  | 55.5  | 62.2  | 70.5  | 75.0  | 78.1  | 78.6  | 75.3  | 68.7  | 57.6  | 49.3 | (6)  |
| (7)  |                                                                   | DBStd       | 12.67 | 4.48  | 5.07  | 6.53  | 7.55  | 6.78  | 4.57  | 3.56  | 3.57  | 4.19  | 5.67  | 5.76  | 4.86 | (7)  |
| (8)  |                                                                   | HDD50       | 291   | 128   | 63    | 18    | 4     | 0     | 0     | 0     | 0     | 0     | 9     | 69    |      | (8)  |
| (9)  |                                                                   | HDD65       | 2226  | 579   | 435   | 306   | 143   | 21    | 0     | 0     | 0     | 26    | 230   | 486   |      | (9)  |
| (10) |                                                                   | CDD50       | 5387  | 14    | 49    | 187   | 369   | 635   | 751   | 872   | 888   | 758   | 579   | 237   | 48   | (10) |
| (11) |                                                                   | CDD65       | 1848  | 0     | 0     | 10    | 59    | 192   | 301   | 407   | 423   | 308   | 140   | 8     | 0    | (11) |
| (12) |                                                                   | CDH74       | 20698 | 0     | 2     | 132   | 791   | 2508  | 3587  | 4622  | 4624  | 3041  | 1324  | 67    | 0    | (12) |
| (13) |                                                                   | CDH80       | 9705  | 0     | 0     | 29    | 270   | 1135  | 1738  | 2366  | 2386  | 1330  | 446   | 5     | 0    | (13) |
| (14) | Wind                                                              | WSAvg       | 5.6   | 5.2   | 5.5   | 6.0   | 6.2   | 6.2   | 6.4   | 7.0   | 6.4   | 5.3   | 4.2   | 4.1   | 4.3  | (14) |
| (15) | Precipitation                                                     | PrecAvg     | 6.1   | 1.3   | 1.2   | 1.1   | 0.4   | 0.1   | 0.0   | 0.0   | 0.0   | 0.0   | 0.3   | 0.6   | 1.1  | (15) |
| (16) |                                                                   | PrecMax     | 11.4  | 3.9   | 3.0   | 2.7   | 2.1   | 0.8   | 0.0   | 0.0   | 0.0   | 0.1   | 1.1   | 3.9   | 3.8  | (16) |
| (17) |                                                                   | PrecMin     | 2.3   | 0.2   | 0.1   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.1  | (17) |
| (18) |                                                                   | PrecStd     | 2.0   | 0.9   | 0.8   | 0.7   | 0.4   | 0.2   | 0.0   | 0.0   | 0.0   | 0.0   | 0.4   | 0.8   | 0.9  | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB    | 64.8  | 72.5  | 84.0  | 92.0  | 99.8  | 99.8  | 102.2 | 102.4 | 98.9  | 93.1  | 80.4  | 69.7 | (19) |
| (20) |                                                                   | 0.4%        | MCWB  | 49.0  | 51.0  | 55.6  | 61.2  | 64.5  | 64.1  | 67.8  | 70.6  | 65.5  | 61.5  | 57.2  | 52.3 | (20) |
| (21) |                                                                   | 2%          | DB    | 61.1  | 67.1  | 77.6  | 86.6  | 94.3  | 96.4  | 96.9  | 96.8  | 94.2  | 88.8  | 75.7  | 65.3 | (21) |
| (22) |                                                                   | 2%          | MCWB  | 47.9  | 49.3  | 54.2  | 58.2  | 62.6  | 64.9  | 67.7  | 68.2  | 64.9  | 61.6  | 56.7  | 50.9 | (22) |
| (23) |                                                                   | 5%          | DB    | 58.4  | 63.1  | 73.0  | 82.5  | 90.3  | 92.7  | 93.5  | 93.9  | 90.6  | 85.2  | 72.1  | 61.6 | (23) |
| (24) |                                                                   | 5%          | MCWB  | 47.4  | 48.3  | 52.0  | 56.7  | 61.4  | 64.3  | 66.7  | 67.2  | 64.5  | 61.3  | 55.1  | 49.9 | (24) |
| (25) |                                                                   | 10%         | DB    | 55.7  | 60.0  | 68.8  | 78.1  | 86.4  | 89.4  | 91.0  | 91.2  | 87.4  | 81.4  | 69.1  | 58.7 | (25) |
| (26) |                                                                   | 10%         | MCWB  | 46.3  | 47.4  | 50.9  | 55.2  | 60.3  | 63.7  | 65.9  | 66.7  | 64.7  | 60.5  | 54.4  | 48.8 | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB    | 52.4  | 54.3  | 59.5  | 63.3  | 67.7  | 71.0  | 73.7  | 75.8  | 73.5  | 68.5  | 62.5  | 56.6 | (27) |
| (28) |                                                                   | 0.4%        | MCDWB | 58.7  | 66.4  | 73.8  | 83.7  | 90.4  | 86.9  | 87.2  | 88.5  | 84.0  | 78.8  | 69.5  | 63.5 | (28) |
| (29) |                                                                   | 2%          | WB    | 50.4  | 51.9  | 56.7  | 60.7  | 64.9  | 68.8  | 71.8  | 73.3  | 70.9  | 66.3  | 60.1  | 53.8 | (29) |
| (30) |                                                                   | 2%          | MCDWB | 56.6  | 60.4  | 70.6  | 78.2  | 87.1  | 84.6  | 86.8  | 84.5  | 81.1  | 77.4  | 68.0  | 60.4 | (30) |
| (31) |                                                                   | 5%          | WB    | 49.1  | 50.3  | 54.6  | 58.7  | 63.4  | 67.3  | 70.3  | 71.9  | 69.2  | 64.8  | 58.2  | 52.1 | (31) |
| (32) |                                                                   | 5%          | MCDWB | 55.3  | 58.8  | 68.0  | 76.2  | 84.1  | 83.1  | 84.7  | 83.7  | 79.9  | 75.7  | 67.1  | 58.4 | (32) |
| (33) |                                                                   | 10%         | WB    | 47.7  | 48.8  | 52.6  | 56.9  | 61.9  | 65.9  | 69.0  | 70.6  | 68.0  | 63.4  | 56.4  | 50.4 | (33) |
| (34) |                                                                   | 10%         | MCDWB | 53.7  | 56.8  | 64.9  | 73.4  | 81.1  | 81.4  | 82.9  | 82.2  | 78.6  | 74.6  | 65.3  | 56.4 | (34) |
| (35) | Mean Daily Temperature Range                                      | 5% DB       | MDBR  | 16.7  | 18.2  | 21.0  | 24.8  | 27.3  | 27.9  | 26.9  | 26.5  | 25.3  | 23.9  | 20.7  | 17.9 | (35) |
| (36) |                                                                   | 5% DB       | MCDBR | 20.9  | 24.0  | 27.5  | 30.3  | 31.1  | 32.0  | 30.0  | 29.6  | 29.0  | 27.9  | 24.8  | 21.6 | (36) |
| (37) |                                                                   | 5% DB       | MCWBR | 11.0  | 10.9  | 11.4  | 12.5  | 12.1  | 11.7  | 10.0  | 9.3   | 9.6   | 10.7  | 12.1  | 12.0 | (37) |
| (38) |                                                                   | 5% WB       | MCDWB | 17.7  | 19.8  | 23.6  | 27.4  | 28.1  | 27.9  | 27.3  | 26.6  | 25.2  | 23.5  | 20.1  | 17.9 | (38) |
| (39) | 5% WB                                                             | MCWBR       | 11.0  | 10.7  | 11.9  | 12.4  | 12.0  | 11.7  | 10.3  | 9.3   | 9.6   | 10.9  | 11.8  | 11.4  | (39) |      |
| (40) | Clear-Sky Solar Irradiance                                        | taub        | 0.342 | 0.371 | 0.413 | 0.457 | 0.458 | 0.387 | 0.393 | 0.399 | 0.392 | 0.412 | 0.364 | 0.344 | (40) |      |
| (41) |                                                                   | taud        | 2.309 | 2.212 | 2.071 | 1.955 | 1.983 | 2.219 | 2.209 | 2.194 | 2.217 | 2.145 | 2.268 | 2.313 | (41) |      |
| (42) |                                                                   | Ebn at Noon | 275   | 278   | 273   | 266   | 266   | 284   | 282   | 279   | 277   | 259   | 264   | 267   | (42) |      |
| (43) |                                                                   | Edh at Noon | 33    | 40    | 50    | 59    | 57    | 45    | 46    | 46    | 43    | 43    | 34    | 31    | (43) |      |
| (44) | All-Sky Solar Radiation                                           | RadAvg      | 953   | 1201  | 1656  | 2018  | 2373  | 2643  | 2556  | 2341  | 1999  | 1512  | 1121  | 892   | (44) |      |
| (45) |                                                                   | RadStd      | 62    | 106   | 98    | 106   | 106   | 35    | 28    | 33    | 49    | 60    | 62    | 56    | (45) |      |

## Historical Trends

|      |                         | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (21 neighbors) | +0.79 | N/A     | -3.11  | N/A   | N/A     | N/A   | N/A | -115        | +284  | +185  |       |

Nomenclature: See separate page